

Semiconductor Technology and Diodes

Sections 19–20

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1 Section 19 — Semiconductor Technology

1.1 Definition

What is a semiconductor?

A **semiconductor** is a kind of material that sits in between a conductor and an insulator in terms of how well it can conduct electricity.

A semiconductor has a conductivity that typically falls within the range of 10^{-7} to 10^3 $(\Omega \cdot \text{cm})^{-1}$. Insulators fall below 10^{-7} $(\Omega \cdot \text{cm})^{-1}$, and conductors have a conductivity greater than 10^3 $(\Omega \cdot \text{cm})^{-1}$.

Where the three material classes sit

$$\underbrace{\sigma < 10^{-7}}_{\text{Insulator}} \quad \underbrace{10^{-7} \leq \sigma \leq 10^3}_{\text{Semiconductor}} \quad \underbrace{\sigma > 10^3}_{\text{Conductor}} \quad (\Omega \cdot \text{cm})^{-1}$$

1.2 Additional Information

Two of the most common semiconductor materials are:

1. Silicon
2. Germanium

1.2.1 Intrinsic Semiconductor

Definition

A pure semiconductor material, as it is, is called an **intrinsic semiconductor**.

The valence shell of a pure semiconductor such as silicon usually has four valence electrons, and these four valence electrons create four covalent bonds with adjacent silicon atoms.

1.2.2 Doping

Definition

It is possible to modify material properties such as the resistivity and conductivity of a semiconductor by intentionally introducing impurities within the pure semiconductor material. This process is known as **doping**. A doped semiconductor material is known as an **extrinsic semiconductor**.

Small amounts of these impurities modify the crystal structure of the semiconductor material, producing a change in the carrier concentration.

Types of Doping.

1. **P-type doping** — if the impurity atom is a *trivalent* atom, like boron (B), then this type of semiconductor material is called a **P-type semiconductor**.
2. **N-type doping** — if the impurity atom is a *pentavalent* atom, like arsenic (As), then this type of semiconductor material is called an **N-type semiconductor**.

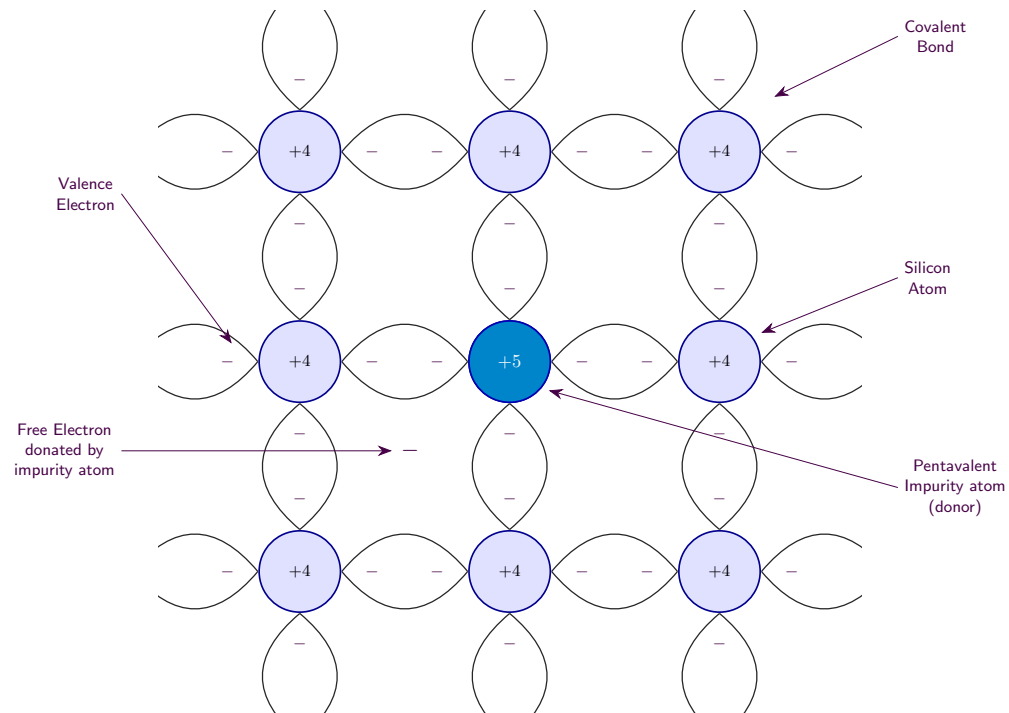


Figure 1: An *n*-type silicon lattice. The pentavalent donor (+5) sits on a silicon site and forms four covalent bonds like its neighbours; its fifth valence electron is left over as a *free electron*.

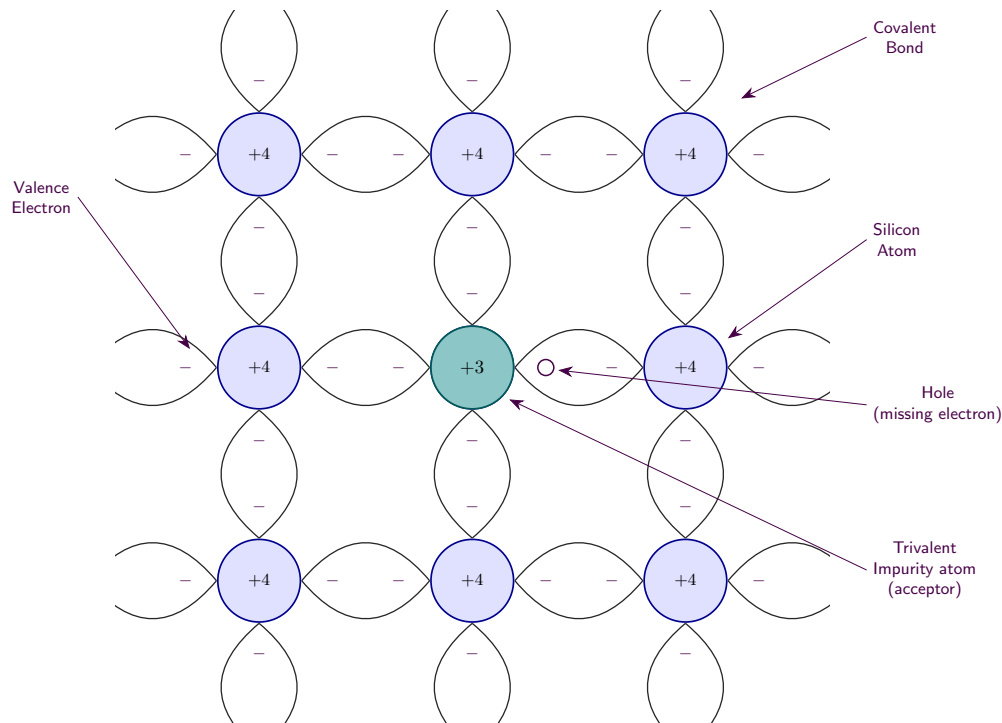


Figure 2: A *p*-type silicon lattice. The trivalent acceptor (+3) can only fill three of its four covalent bonds; the vacant fourth site is a *hole*.

Substitution Mechanism.

- A dopant atom substitutes a single silicon atom in the crystal lattice.
- Depending on the dopant type, it contributes:
 - an extra **hole** (absence of an electron) — for a P-type impurity;
 - an extra **electron** — for an N-type impurity.
- These holes and electrons act as the **charge carriers** in P-type and N-type semiconductors, respectively.

Pentavalent dopant → N-type → mobile electrons are the carriers
 Trivalent dopant → P-type → holes are the carriers

Charge Neutrality and Carrier Mobility.

- The impurity concentration is much lower than the density of the pure semiconductor atoms (such as silicon), so only a small fraction of the atoms are actually replaced.
- A dopant atom is electrically neutral, just like the atoms of the intrinsic semiconductor itself — doping does not introduce any additional net charge into the material.
- The extra electron or hole contributed by the dopant atom is mobile, free from the influence of any single silicon or dopant atom's nucleus.
- These dopant-contributed carriers are therefore the charge carriers in the resulting **extrinsic semiconductor** material.

1.3 Energy Band Diagrams

The way a semiconductor material behaves can be explained through energy band diagrams.

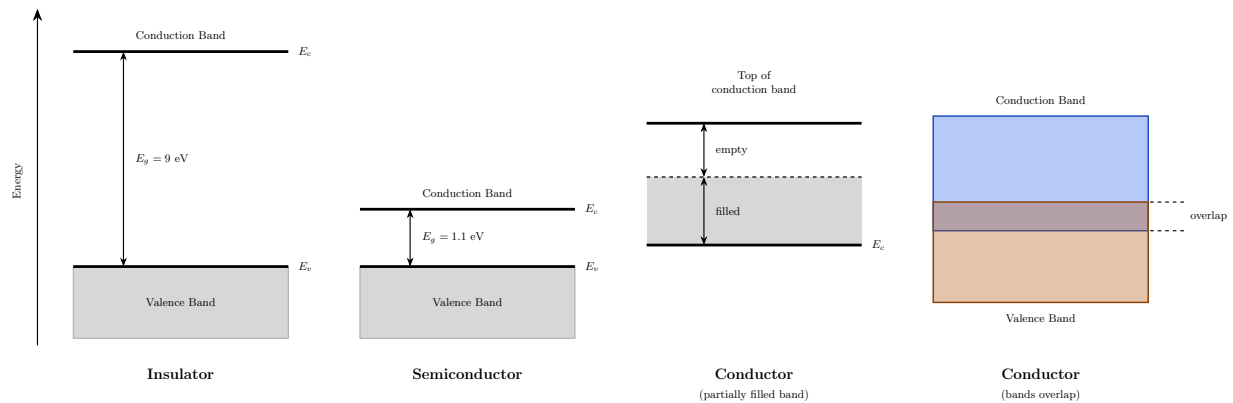


Figure 3: Energy band diagrams of an insulator, a semiconductor and a conductor. The two conductor panels show the two ways a material can end up with no band gap to cross.

The three energy ranges

- **Conduction band** — the range of energies within which electrons can flow freely throughout the material.
- **Band gap** — immediately below the conduction band, a range of energies within which electrons are *not permitted* to exist, because of the crystal structure of the material.
- **Valence band** — below the band gap. Electrons with energies in this band are bound to the atomic cores (nuclei) within the crystal structure, rather than being free to conduct.

From Figure 3, the gap between the electrons bound to the material (valence band) and the electrons that are free to conduct through it (conduction band) is greatest in **insulators** — around 9 eV.

In **conductors**, by contrast, there is no such gap between the valence band and the conduction band. This can happen in either of two ways, and the last two panels of Figure 3 show both:

1. **Partially filled conduction band** — the conduction band itself is only part-full, so the electrons occupying the filled portion can move into the empty states directly above them without having to cross any band gap.
2. **Overlapping bands** — the minimum of the conduction band energy lies *below* the maximum of the valence band energy, so the two bands overlap. An electron at the top of the valence band is already at an energy that belongs to the conduction band, and can conduct without any excitation at all.

Either way there is no forbidden range of energies to cross, which is why conductors conduct so readily.

In **semiconductors**, however, the band gap between the valence band and the conduction band is much smaller — around 1.1 eV.

Band gap comparison

$$\underbrace{E_g \approx 9 \text{ eV}}_{\text{Insulator}}$$

$$\underbrace{E_g \approx 1.1 \text{ eV}}_{\text{Semiconductor}}$$

$$\underbrace{E_g = 0 \text{ (no gap)}}_{\text{Conductor}}$$

2 Section 20 — Diodes

2.1 Definition

Definition

The most simple application of semiconductor materials occurs in **diodes**. A diode is a two terminal semiconductor device that has a P-type region in tandem with an N-type region.

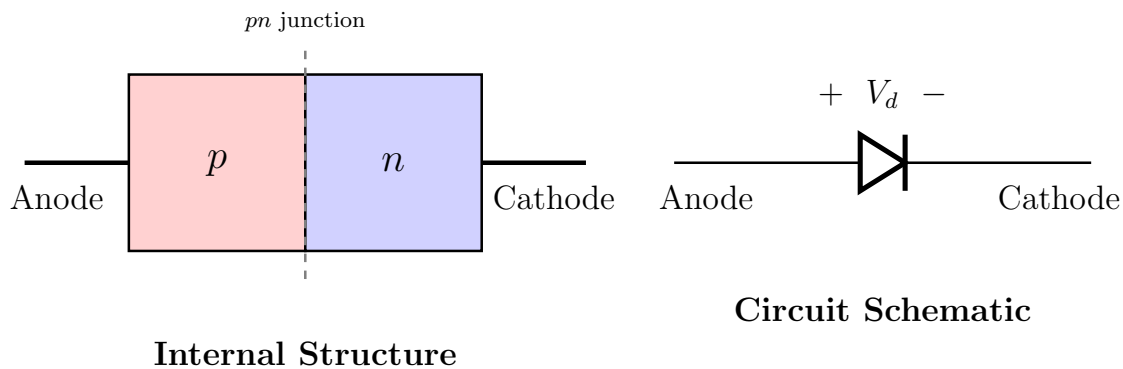


Figure 4: The internal structure of a *pn* junction diode (left) and its circuit schematic (right).

The two terminals of a diode are the **anode** (on the P-type side) and the **cathode** (on the N-type side).

The diode as a one-way gate

A diode is a two-lead semiconductor device that acts as a **one-way gate** to electric current flow.

- **Forward biasing** — the anode is made *more positive* than the cathode. Current is permitted to flow through the device.
- **Reverse biasing** — the anode is made *more negative* than the cathode. The diode blocks current flow.

2.1.1 Applications

Diodes are used most commonly in circuits that convert AC voltages and currents into DC voltages and currents, for example an AC/DC power supply. Beyond rectification they also appear in:

- Voltage-multiplier circuits
- Voltage-shifting circuits
- Voltage-limiting circuits
- Voltage-regulator circuits

2.1.2 Construction

A *pn* junction diode is also called a **rectifier diode**. It is formed by sandwiching together n-type and p-type silicon:

- In practice, manufacturers grow an n-type silicon crystal and then abruptly change it to a p-type crystal.
- A glass or plastic coating is then placed around the joined crystals.
- The n side is the **cathode** end and the p side is the **anode** end.

Both n-type and p-type silicon conduct electric current — one with electrons, the other with holes. What makes the device a one-way gate is the way these two kinds of carrier interact with each other, and with the electric field supplied by an external voltage across the leads.

2.2 Additional Information

2.2.1 How a PN Junction Works

When a P-type semiconductor material is placed side by side with an N-type material, then at the interface between the two, holes from the P-type material flow into the N-type material up to a certain region, making the region close to the PN interface within the N-type material positively charged.

Similarly, electrons from the N-type region can seep into the P-type region, creating a negatively charged layer close to the interface within the P-type material.

The space charge region

This separation of positive and negative charges at the interface of the PN junction creates a region called the **space charge region**, which acts as a barrier to current flow.

2.2.2 Modes of Operation

The width of the depletion region is not fixed — an externally applied voltage either squeezes it or widens it, and this is what gives the diode its one-way behaviour. A diode therefore has two modes of operation.

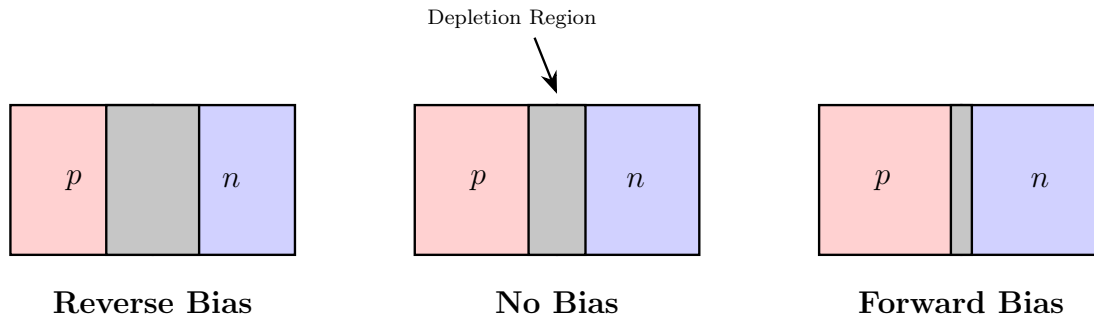


Figure 5: The depletion region under reverse bias, no bias and forward bias. Reverse bias widens the barrier; forward bias narrows it.

The two modes

Forward Bias (FB)

- The depletion region is **constricted**.
- Allows electrons to flow through the junction.
- Ideally acts as a **short circuit**.

Reverse Bias (RB)

- The depletion region is **expanded**.
- Bars (does not allow) electron flow through the junction.
- Ideally acts as an **open circuit**.

Forward Bias → depletion region shrinks → current flows → short circuit
 Reverse Bias → depletion region grows → no current → open circuit

What the carriers actually do.

- **Forward bias** — electrons from the n side and holes from the p side are forced *toward* the centre (the *pn* interface) by the electric field supplied by the battery. The electrons and holes combine, and current passes through the diode.
- **Reverse bias** — holes on the n side and electrons on the p side are pulled *away* from the junction. This leaves an empty zone around the *pn* junction that is free of charge carriers — the **depletion region**. This region has an insulative quality that prevents current from flowing.

2.2.3 Threshold Voltage

A diode's one-way gate feature does not work all the time. It takes a minimal voltage to turn the diode on even when it is forward biased.

Junction threshold voltage

Current flows from anode (+) to cathode (−) only once the **forward voltage** V_F across the diode exceeds the **junction threshold voltage**. Below that voltage the diode does not conduct.

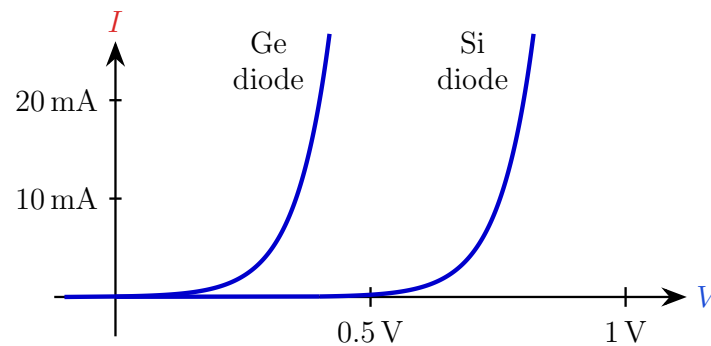


Figure 6: Current versus voltage for silicon and germanium diodes. The germanium diode begins conducting at a noticeably lower voltage than the silicon diode.

Table 1: Rule-of-thumb junction threshold voltages, and the spread actually seen in real components.

Diode type	Rule of thumb	Real-life range
Silicon <i>pn</i> junction	0.6 V	0.6 – 1.7 V
Germanium	0.2 V	0.2 – 0.4 V
Schottky	0.4 V	0.15 – 0.9 V

Do not take the rule-of-thumb column too seriously — with real components these thresholds may be a few tenths of a volt off.

Requiring a specific voltage to turn on may seem like a drawback, but it is in fact very useful: it makes the diode a **voltage-sensitive switch**.

Silicon versus germanium. Besides the forward-biasing voltage, the two materials differ in their ability to dissipate heat.

- Silicon diodes do a better job of dissipating heat than germanium diodes.
- When germanium diodes exceed 85°C, thermal vibrations affect the physics inside the crystalline structure to the point where normal diode operation becomes unreliable. Above 85°C, germanium diodes become worthless.

2.2.4 Ratings and Limits

Two ratings you must not exceed

- **Peak current rating** $I_{0(\max)}$ — do not supply a forward current I_F beyond it, or the junction melts down internally.
- **Peak inverse voltage (PIV) rating** — do not apply a reverse voltage V_R any larger than this, or the diode is destroyed.

Note that putting a large voltage straight across a forward-biased diode makes a very large current flow, which will probably destroy it. A diode is almost always used with a series resistor to limit that current.

2.3 Kinds of Rectifiers / Diodes

There are numerous types of diodes, each designed to work better in one application or another.

Table 2: Common diode families and what each is designed for.

Family	Designed for
Rectifier diodes	High-power applications (switching, power supplies) that draw large currents or rectify high voltages.
Signal / switching / fast recovery / high speed	Low internal capacitance — they store less charge but usually have weaker junctions for large currents. At high speeds this reduces RC switching time constants, meaning fewer time delays and lower signal losses.
Schottky diodes	Particularly low junction capacitance and fast switching (~ 10 ns), thanks to a metal–semiconductor junction interface, plus a low threshold voltage.
Germanium diodes	RF signal detection and low-level logic design, using their small ≈ 0.2 V threshold.

Why Schottky diodes are so widely used.

- Their low threshold lets them detect low-voltage, high-frequency signals that ordinary pn junction diodes would not see. A Schottky with a 0.3 V threshold passes signals greater than 0.3 V, whereas a silicon pn junction diode with a 0.7 V threshold only passes signals greater than 0.7 V.
- This makes them popular in low-voltage signal rectifiers in RF circuits, signal switching in telecommunications, small DC/DC converters, small low-voltage power supplies, protection circuits and voltage clamping arrangements.
- Their high current density and low voltage drop also make them excellent in power supplies, since they generate less heat and therefore need smaller heat sinks.

Germanium diodes, by contrast, are not used in high-current rectifying applications, since they are weaker and leak more than silicon.

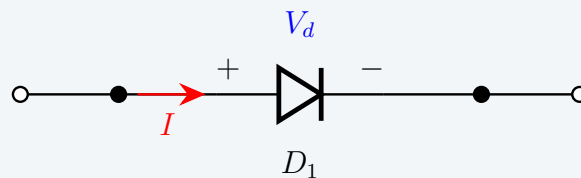
3 Diode Circuit Models

A diode can be represented by several circuit models. They trade off *simplicity* against *accuracy*. Only the two extremes of that range are covered here.

Model	Complexity	Accuracy
Ideal Diode Model	Simple	Low
Exponential (Shockley) Model	More complex	Increased accuracy

The ideal model is a **piecewise linear I–V model**; the Shockley model is the real (exponential) characteristic.

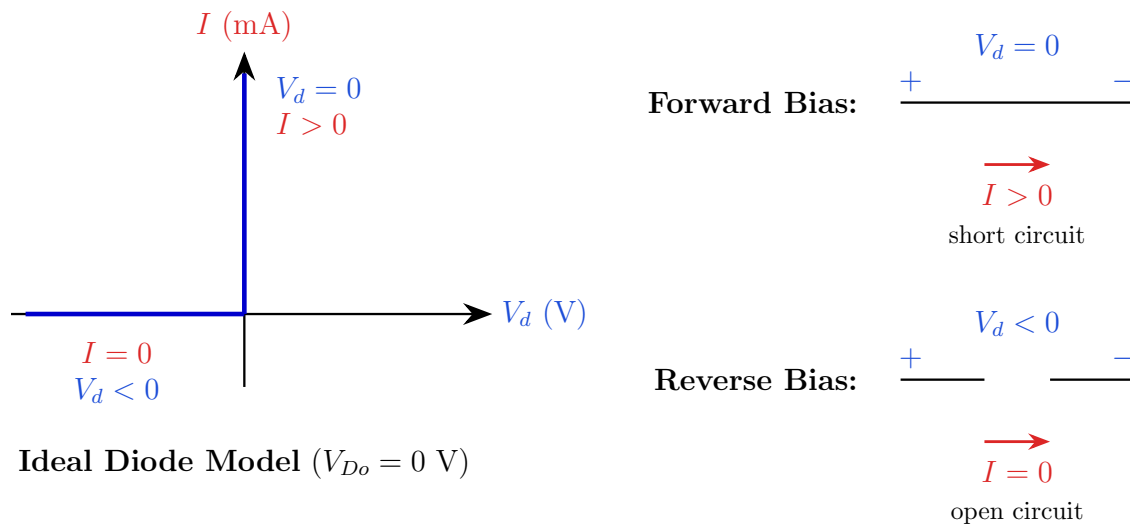
Notation used across the models



The sign convention above fixes what every symbol below means: V_d is measured + at the anode and – at the cathode, and I is taken positive flowing *into* the anode, i.e. in the direction the diode conducts.

V_d	voltage across the diode
I	current through the diode (anode to cathode)
V_{Do}	diode cut-off voltage
I_s	reverse saturation current (Shockley model)

3.1 The Ideal Diode Model



The ideal diode model uses a cut-off voltage of $V_{Do} = 0$ V, so the diode conducts as soon as the voltage across it becomes positive.

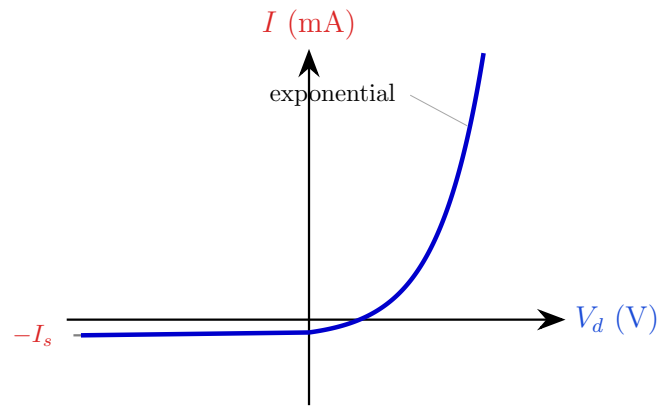
Ideal diode model

$\underbrace{I > 0, V_d = 0}_{\text{Forward Bias} \Rightarrow \text{short circuit}}$

$\underbrace{I = 0, V_d < 0}_{\text{Reverse Bias} \Rightarrow \text{open circuit}}$

3.2 The Shockley (Exponential) Model

The ideal model above is a piecewise linear approximation. The **Shockley model** is the real, continuously curved characteristic of a *pn* junction.



Shockley (Exponential) Model — the real diode characteristic

In terms of fundamental constants, the current through a *pn* junction is

$$I = I_0 \left(e^{qV_d/(nkT)} - 1 \right)$$

where q is the elementary charge, k is Boltzmann's constant and T is the absolute temperature. Collecting kT/q into a single quantity — the **thermal voltage** V_T — gives the working form of the equation. (The saturation current is written I_0 or I_s ; both denote the same quantity.)

Shockley diode equation

$$I = I_s \left(e^{V_d/(nV_T)} - 1 \right) \quad V_T = \frac{kT}{q} \approx 25.9 \text{ mV at } 300 \text{ K}$$

I diode current

I_s **reverse saturation current** — very small, e.g. 10^{-12} A

V_d applied voltage across the diode

n ideality factor (≈ 1 for silicon)

In hand calculations the rounded value $V_T \approx 0.026$ V is used.

There is no separate “forward” and “reverse” case here — the single equation covers both. For V_d negative and a few V_T in magnitude, the exponential term vanishes and $I \rightarrow -I_s$, which is the small reverse leakage current.

3.3 Using the Diode Equation

Problem

For a silicon diode with $I_s = 10^{-12}$ A, $n = 1$ and $V_T = 0.026$ V, evaluate the diode current I at $V_d = 0.7$ V, 0.6 V, -5 V and 0.1 V.

Forward bias at $V_d = 0.7$ V

$$\frac{V_d}{nV_T} = \frac{0.7}{0.026} \approx 26.9 \quad \Rightarrow \quad e^{26.9} \approx 4.8 \times 10^{11}$$

$$I = 10^{-12} (e^{26.9} - 1) \approx 10^{-12} \times 4.8 \times 10^{11} \approx 0.48 \text{ A} \approx 0.5 \text{ A}$$

Forward bias at $V_d = 0.6$ V

$$\frac{0.6}{0.026} \approx 23.1 \quad e^{23.1} \approx 1.1 \times 10^{10} \quad I \approx 10^{-12} \times 1.1 \times 10^{10} = 0.011 \text{ A} = 11 \text{ mA}$$

Reverse bias at $V_d = -5$ V

$$\frac{-5}{0.026} \approx -192 \quad e^{-192} \approx 0 \quad I \approx 10^{-12} \times (0 - 1) = -10^{-12} \text{ A} = -I_s$$

Small forward voltage $V_d = 0.1$ V

$$\frac{0.1}{0.026} \approx 3.85 \quad e^{3.85} \approx 47 \quad I \approx 10^{-12} \times (47 - 1) \approx 4.6 \times 10^{-11} \text{ A}$$

What the four evaluations teach

- **The exponential is savage** — adding just 0.1 V (from 0.6 to 0.7 V) multiplies the current by $e^{0.1/0.026} \approx 47$: from 11 mA to ≈ 0.5 A.
- **Reverse bias saturates** — beyond a few V_T of reverse voltage the current pins at the tiny $-I_s$, whatever the voltage. This is the flat reverse branch of the characteristic.
- **Small forward voltages conduct almost nothing** — at 0.1 V the current is only tens of picoamps.
- Together these are exactly why the piecewise models work: below ≈ 0.6 V the diode is effectively off, and beyond it the current climbs so steeply that the diode voltage barely moves — which is where the CVD model's $V_{Do} \approx 0.6$ – 0.7 V comes from.

3.4 Summary of the Ideal Diode Model

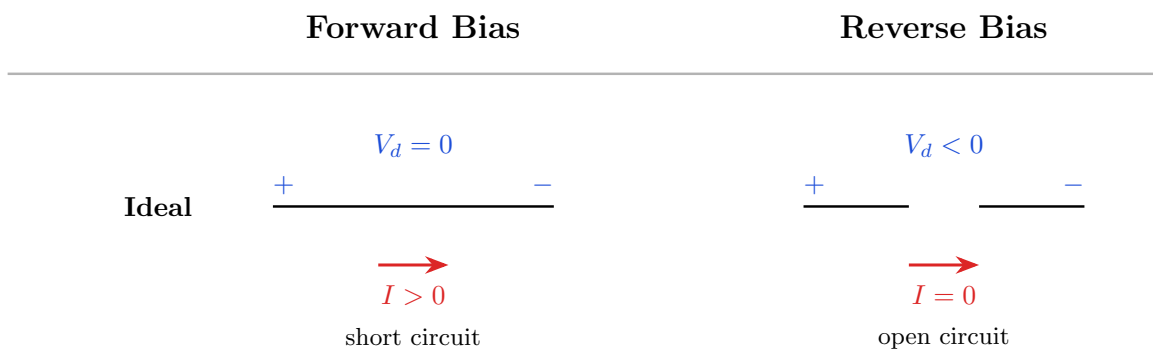


Figure 7: Equivalent circuits for the ideal diode model in each bias mode. Forward bias is a short circuit and reverse bias an open circuit. The Shockley model has no such piecewise equivalent circuit — it is used through its equation directly.

4 Solving Circuits with Diodes

A diode is a non-linear element, so we do not know in advance whether it is conducting. This is resolved with the **Method of Assumed States**, which has only *two* states to choose between.

The Method of Assumed States for diodes

1. **Assume** — assume one of the two operating modes, Forward Biased (FB) or Reverse Biased (RB). *A correct assumption makes solving the circuit easier.*
2. **Solve** — use the corresponding equivalent circuit with KCL/KVL or any circuit analysis tool of your choice.
3. **Verify** — check whether the conditions of the chosen mode are satisfied. If not, one can be sure that *the other* operating mode is the correct one.

Assume FB or RB $\rightarrow$ Solve $\rightarrow$ Verify $\rightarrow$ (if it fails, the other mode must be correct)

Because there are only two modes, a failed verification immediately identifies the answer.

The conditions to check for the ideal model are:

Table 3: Verification conditions for the ideal diode model.

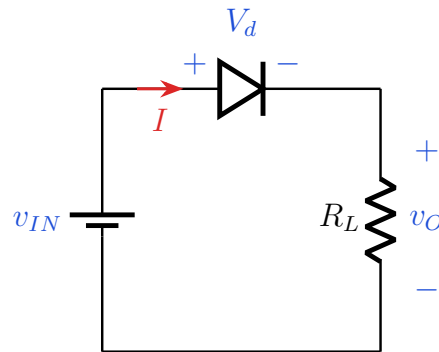
Model	Forward Bias (FB)	Reverse Bias (RB)
Ideal	$V_d = 0, I > 0$	$I = 0, V_d < 0$

Both examples that follow use the same basic circuit: a source v_{IN} driving a diode in series with a load resistor $R_L = 2.5 \text{ k}\Omega$, with the output v_O taken across R_L .

4.1 Example 1 — Ideal Diode, Assumption Correct

Problem

For the circuit below with $R_L = 2.5 \text{ k}\Omega$ and $v_{IN} = 5 \text{ V}$, find v_O , V_d and I . Treat the diode as ideal.



Assume: The diode is in **FB**, i.e. a short circuit, where

$$V_d = 0 \text{ V} \quad I > 0 \text{ mA}$$

Solve: With the diode shorted, the source appears directly across R_L :

$$\begin{aligned} v_O &= v_{IN} = 5 \text{ V} \\ V_d &= v_{IN} - v_O = 0 \text{ V} \\ I &= \frac{v_O}{R_L} = \frac{5 \text{ V}}{2.5 \text{ k}\Omega} = 2 \text{ mA} > 0 \end{aligned}$$

Verify: For FB we require $V_d = 0 \text{ V}$ and $I > 0 \text{ mA}$. Both conditions are fulfilled.

Assumption is CORRECT!

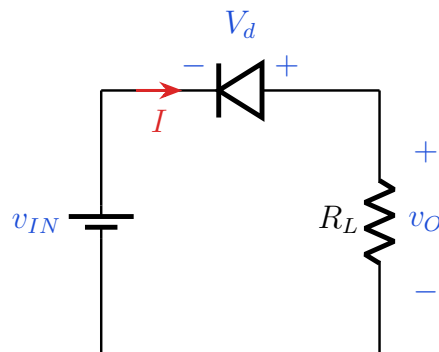
Answers

$$v_O = 5 \text{ V} \quad V_d = 0 \text{ V} \quad I = 2 \text{ mA}$$

4.2 Example 2 — Ideal Diode, Assumption Rejected

Problem

The same circuit and the same $v_{IN} = 5\text{ V}$, but the diode is now **reversed**. Find v_O , V_d and I , treating the diode as ideal.



First Attempt

Assume: The diode is in **FB**, i.e. a short circuit, where

$$V_d = 0\text{ V} \quad I > 0\text{ mA}$$

Solve:

$$v_O = v_{IN} = 5\text{ V}$$

$$V_d = v_O - v_{IN} = 0\text{ V}$$

$$I = -\frac{v_O}{R_L} = -\frac{5\text{ V}}{2.5\text{ k}\Omega} = -2\text{ mA} < 0$$

Verify: FB requires $I > 0\text{ mA}$, but here $I = -2\text{ mA} < 0$.

Assumption is INCORRECT!

The diode must be reverse biased.

Second Attempt

Assume: The diode is in **RB**, i.e. an open circuit, where

$$V_d < 0 \text{ V} \quad I = 0 \text{ mA}$$

Solve: With no current flowing there is no drop across R_L :

$$\begin{aligned} I &= 0 \text{ mA} \\ v_O &= IR_L = 0 \text{ V} \\ V_d &= v_O - v_{IN} = -5 \text{ V} < 0 \text{ V} \end{aligned}$$

Verify: For RB we require $I = 0$ and $V_d < 0 \text{ V}$. Both conditions are fulfilled.

Assumption is CORRECT!

Answers

$$v_O = 0 \text{ V} \quad V_d = -5 \text{ V} \quad I = 0 \text{ mA}$$

4.3 Example 3 — A Clamp Circuit

Problem

What does the output look like for circuit (a) below? What happens if a load of $2.2 \text{ k}\Omega$ is attached to the output, as in circuit (b)?

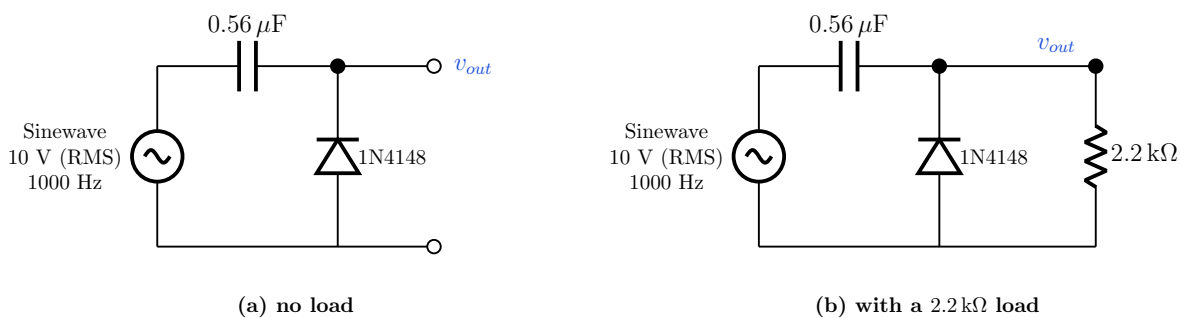


Figure 8: A diode clamp circuit driven by a 10 V RMS, 1000 Hz sinewave, shown without a load (a) and with a $2.2 \text{ k}\Omega$ load (b).

Without a load

This is a **clamp circuit**: the output is shifted so that it is practically pure alternating DC, with the exception of a 0.6 V negative dip due to the diode drop.

Recalling that $V_{\text{peak}} = 1.41 \times V_{\text{rms}}$,

$$V_{\text{peak}} = 1.41 \times 10 \text{ V} = 14.1 \text{ V}$$

so the waveform swings over a $2V_{\text{peak}}$ range sitting just above the clamp level.

Answer — circuit (a)

$$\text{maximum} = 27.6 \text{ V} \quad \text{minimum} = -0.6 \text{ V}$$

The diode holds the bottom of the waveform at one diode drop below ground, and the capacitor shifts the whole sinewave up by that amount.

With the 2.2 k Ω load attached

With the load attached, the output level decreases slightly. The capacitor and load resistor together act like a **high-pass filter**, with a cut-off frequency of

$$f_c = \frac{1}{2\pi RC}$$

Answer — circuit (b)

In simulation the output falls to 8.90 V (RMS), i.e. 24.5 V peak, with a -0.6 V minimum.